MILITARY SPECIFICATION

SWITCHING GROUP, TRANSMITTER ANTENNA OK-261/BRC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, development, performance, and test requirements for the remotely controlled transmitter antenna switching matrix group (hereinafter referred to as the equipment) consisting of a radio frequency (RF) switch matrix and a remote control unit. The matrix is used to interconnect transmitters and antennas in a submarine communications system and shall operate over a frequency range of direct current (DC) to 400 megahertz (MHz).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue listed in the Department of Defense Index of Specifications and Standards (DoDISS) and its supplements, form a part of this document to the extent specified herein. The date of the applicable DoDISS and Supplements thereto shall be as specified in the solicitation.

SPECIFICATIONS

MILITARY

MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-S-22885	Switch, Push Button, Illuminated, General Specification For
MIL-C-24308	Connector, Electric, Rectangular, Miniature Polarized Shell, Rack And Panel, General Specification For
MIL-C-26777	Cable Assembly, Electronic Test Equipment,(3 Wires, 125 And 250 Volts AC And 28 Volts DC) Grounding Plug Connector, General Specification For
MIL-C-39012	Connector, Coaxial, Radio Frequency, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-C-55302	Connector, Printed Circuit Subassembly And Accessories

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-189	Rack, Electrical Equipment, 19-Inch And Associated Panels
MIL-STD-275	Printed Wiring For Electronic Equipment
MIL-STD-454	Standard General Requirement For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Demonstration
MIL-STD-740	Airborne And Structureborne Noise Measurements And Acceptance Criteria Of Shipboard Equipment
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1378	Requirements For Employing Standard Electronic Modules
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The equipment shall conform to the requirements of MIL-E-16400, except as hereinafter specified.

3.1.1 First article. When specified (see 6.2.1) the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

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3.1.2 Equipment description. The equipment shall be designed in accordance with the requirements specified herein to provide the RF communication switching functions for interconnection of high frequency (HF) and ultra-high frequency (UHF) radio transmitters and antennas on board 688 Class SSNs. The switching matrix group shall consist of the following:

a. One coaxial type of transmission line RF switch matrix (Unit 1)
b. One remote control (Unit 2)

3.1.2.1 RF switch matrix. The transmission stripline RF switch matrix (Unit 1) shall provide the RF, interlock, and control switching functions necessary for the interconnection of two HF transmitters, one spare HF transmitter, two UHF transceivers, and one antenna test set with three HF antennas, two UHF antennas, one dummy load (HF and UHF), and spare outputs for two UHF antennas and one HF antenna. Primary control of the switch functions shall be supplied from the remote control (Unit 2), with local manual back-up switching of the RF paths by means of a front panel knob.

3.1.2.2 Remote control unit. The remote control unit shall consist of a six-row, five-column pushbutton matrix which duplicates the configuration of the RF switch matrix as specified in 3.2.1.11. The operation of a pushbutton shall energize the corresponding actuator assembly in the RF switch matrix, rotating the rotor assembly to the TURN position and establishing the desired transmitter and antenna RF path. Also, through an electronic interlock circuit all other crosspoints in the same row and column that are in the TURN position shall be set to the THRU position. Indicator lamps in the pushbuttons shall indicate the actual switch status and shall verify the occurrence of switching. Labeling of inputs and outputs shall be consistent with the requirements specified in 3.2.1.11.

3.2 Characteristics.

3.2.1 Performance characteristics.

3.2.1.1 Frequency range. The RF switch matrix shall be capable of operating over a frequency range of DC to 400 MHz.

3.2.1.2 Average power rating. Switches used in the RF switch matrix shall be capable of continuous operation at 2 kilowatts (kW) continuous wave (CW) power at 2 to 30 MHz and 400 watts (W) at 225 to 400 MHz with a maximum voltage standing wave ratio (VSWR) of 1.2:1.

3.2.1.3 Peak power rating. The RF switches shall be capable of handling at least 5 kW peak power at 2 to 30 MHz and 1 kW peak power at 225 to 400 MHz at a VSWR of up to 4:1.

3.2.1.4 Insertion loss. The insertion loss through any path in a single switch shall be less than 0.1 decibel (dB) from DC to 30 MHz and less than 0.2 dB from 30 to 400 MHz.

3.2.1.5 Characteristic impedance. The nominal characteristic impedance of each RF circuit shall be 50 ohms.

3.2.1.6 Crosstalk. The transmit mode crosstalk between any two valid switch contacts when terminated in the characteristic impedance shall be down at least 75 dB when measured at 30 MHz, and 50 dB when measured at 400 MHz.

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3.2.1.7 Isolation. Isolation between the active switch crosspoints and any other open position shall not be less than 75 dB at 30 MHz and 50 dB when measured at 400 MHz.

3.2.1.8 Intermodulation. The third harmonic of a fundamental tone shall be at least 100 dB below the fundamental.

3.2.1.9 VSWR. The VSWR shall not exceed 1.1:1 from 2 to 30 MHz and 1.2:1 from 30 to 400 MHz.

3.2.1.10 Functional performance. The RF switch matrix shall provide for a stripline transmission line assembly arrangement with rear vertical transmission lines as columns and the front horizontal transmission lines as rows. A rotor assembly at each column and row intersection shall be provided to form a 5 by 6 switching matrix. Actuator assemblies shall be mounted on the rear of the switch so as to engage the switch rotor assembly and shall be controlled by a remote control unit. The rotary assembly shall be designed to permit manual operation of the motorized crosspoints by means of a front panel knob.

3.2.1.11 Split matrix design. The stripline matrix switch shall be a 5 by 6 split matrix design configured as shown in FIGURE 1. The transmitter and antenna designations shall be engraved on removable strips mounted on a subpanel visible through the front panel. The RF paths shall be indicated by arrows silk-screened on this subpanel.

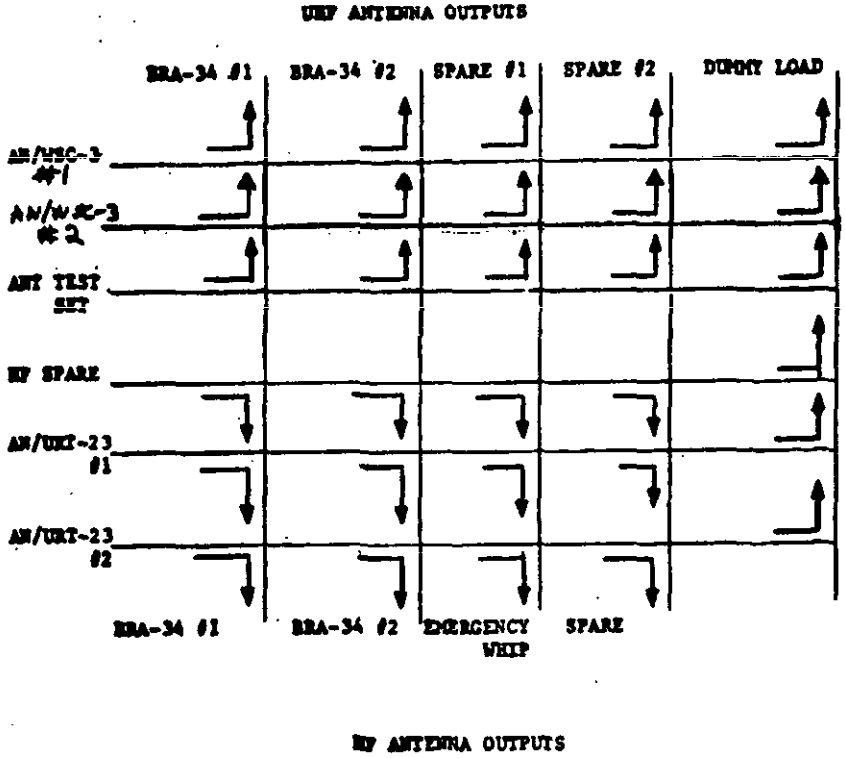


FIGURE 1. Configuration for split matrix.

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3.2.1.12 Matrix switch reconfiguration. The switch matrix shall be capable of being reconfigured into a full matrix or any other practical split matrix configuration by an appropriate field change modification.

3.2.1.13 Interlocks. The RF switch matrix shall provide interlock and bus circuits for each RF input and output path in accordance with the requirements of FIGURE 2. Two interlock circuits shall be required for each input to each output and they shall duplicate the switching of the RF circuit to provide complete interlock. Four additional separate bus circuits shall be provided between UHF transmitter and UHF antenna connectors. Multipin connectors shall be provided on the rear of the RF switch matrix to provide input and output interlock interface to transmitters, antennas, and other units as shown in FIGURE 2. The interlock circuits shall open at least 20 milliseconds (ms) prior to RF switch opening, and RF switch opening, and RF switch closure shall occur within 50 ms before interlock circuit closure.

3.2.1.14 Pushbutton characteristics. Pushbutton switches shall conform to the requirements of MIL-S-22885. The lens color of matrix switches shall be green and shall be lighted upon connection of the activated RF path. The lamps to be utilized within the pushbutton matrix shall achieve a long life status by operating at 90 percent of rated voltage and shall be easily replaced from the front of the pushbutton switch matrix without special tools. Lens size and shape shall be a minimum of 1.25 centimeters (cm) (0.50 inch (in)) by 1.25 cm (0.50 in) square. Dual indicator lamps for each matrix pushbutton switch shall be used in lieu of a lamp test switch. A status indicator lamp and a lamp test switch shall be provided to permit verification of equipment status.

3.2.1.15 Remote clear pushbutton. Each row of the 5 by 6 remote pushbutton switch matrix shall be provided with an unlighted clear pushbutton to be labeled CLEAR. The function of the CLEAR pushbutton shall be to provide for all RF switch matrix switches in that row to be in the THRU position.

3.2.1.16 Primary power source. The equipment shall be designed to operate from a primary power source of 115 volts alternating current (VAC) ± 10 percent, 60 hertz (Hz) ± 5 percent single-phase. The total power required to operate the equipment shall not exceed 175 W.

3.2.1.16.1 Power cable assembly. The input power cable assembly shall be Type I of MIL-C-28777. Power cable connection to the equipment shall be from the rear.

3.2.2 Physical characteristics.

3.2.2.1 Dimensions and weights. The maximum dimensions of the equipment shall not exceed the dimensions and weights shown in TABLE I.

TABLE I. Maximum dimensions and weights.

Item	Width (cm)	Width (in)	Depth (cm)	Depth (in)	Height (cm)	Height (in)	Weight Kilograms (Kg)	Weight Pounds (lb)
RF matrix	48.26	19	45	18	30.5	12.25	29.25	65
Control unit	48.26	19	45	18	17.5	7	15.75	35

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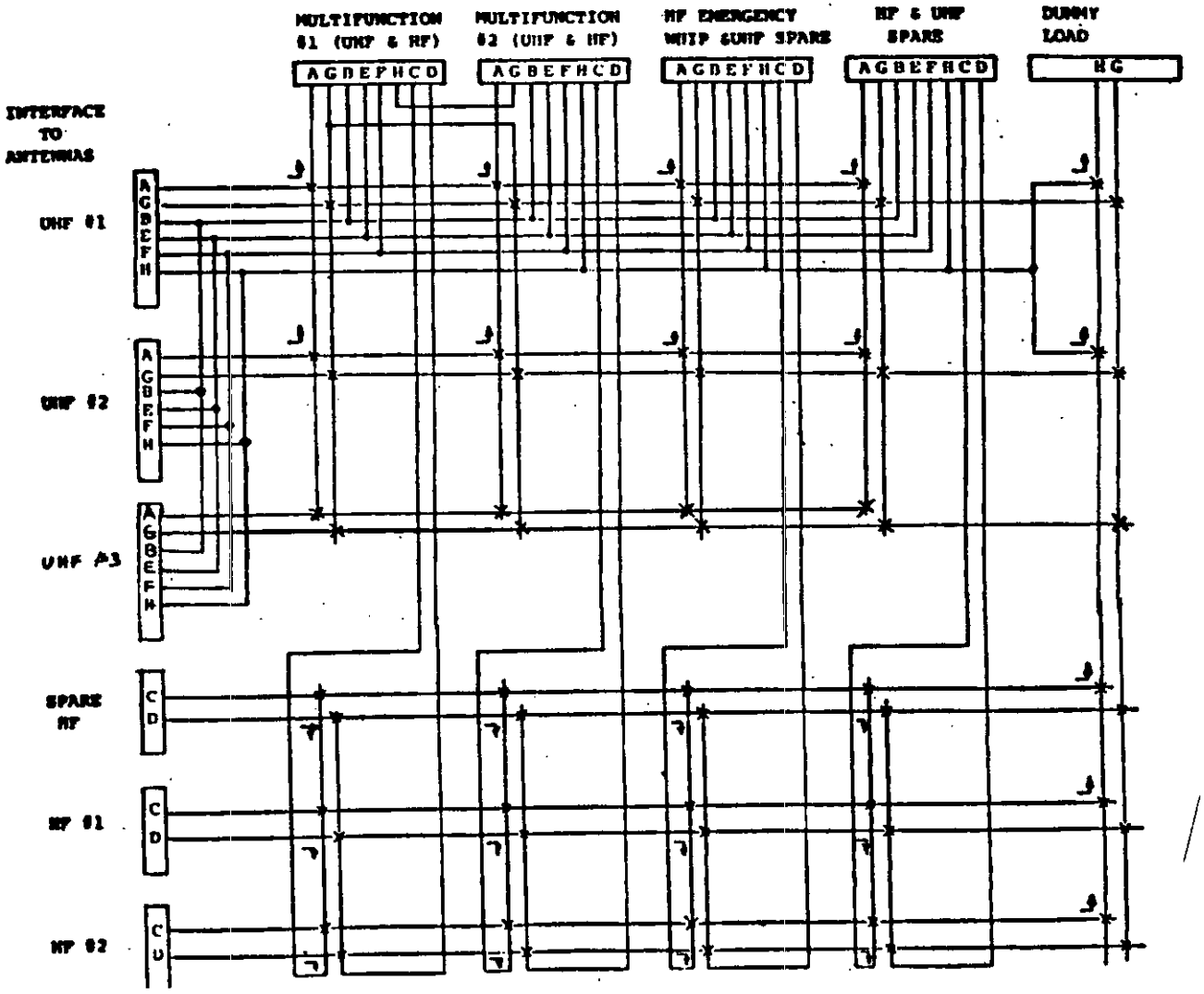


FIGURE 2. Switch matrix interlock control switching.

MIL-C-28854(EC)

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DESCRIPTION OF SIGNAL AND INTERLOCK

Connector pin	Function
A	Plus 28 VDC originating in either multifunction or spare antenna to be switched to selected UHF transmitter SATCOM MODE CLOSED CONTACTS. Not required for dummy load.
B	Return bus for signal A to multifunction or spare antenna unit. Not required for dummy load.
C and D	HF keyline interlock signal and return, originating in either HF transmitter or HF spare with both sides switched to selected antenna or dummy load connectors for total isolation between transmitters (see note).
E and F	Plus 12 VDC in transmit signal and return, originating in selected UHF transmitter for activation of relays in both multifunction antennas in parallel. No switching required and not required for dummy load.
G	UHF keyline interlock signal, origination in selected UHF transmitter to be switched in series to both multifunction antennas when either is selected, or switched separately to spare UHF antennas or dummy load.
H	Return for signal G from antenna or dummy load to transmitter. No switching required for antenna connection but switching is required for dummy load connection (see note).

NOTE: At the dummy load connector, pins G and H will be shared by both HF and UHF keyline interlock functions but shall be completely isolated by internal switching on each line.

FIGURE 2. Switch matrix interlock control switching. (Continued)

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3.2.2.2 Protective cover. A protective cover shall be provided to protect the front panel manual override controls of the RF switch matrix and the rear-mounted wafer switch motor actuator assemblies. The design of both covers shall provide easy access and removal.

3.2.2.3 Interface connector location. Interface connectors shall be located on the rear plane of each unit. Interfacing RF connectors shall be placed a minimum of 5 cm (2 in) from each other and from external packaging protrusions so that external RF cables with conduit adapters of 5 cm (2 in) in diameter may be connected.

3.2.2.4 Maintenance configuration. The RF switch shall be designed to be capable of being moved outward on its slides for purposes of trouble-shooting, maintenance, and reconfiguration of antenna and transmitter connections. The RF switch matrix shall be capable of being fully operational when pulled out for maintenance.

3.2.3 Reliability.

3.2.3.1 Quantitative reliability requirements. The upper test mean-time-between-failures (MTBF) (θ_0 as defined by MIL-STD-781) shall be 2200 hours. The lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 1100 hours. Individual RF switch assemblies shall have a life expectancy of at least 50,000 cycles without maintenance or overhaul.

3.2.4 Maintainability.

3.2.4.1 Quantitative corrective maintenance. The equipment shall have a mean-corrective-maintenance-time (M_{ct}) not exceeding 0.5 hours and a maximum-corrective-maintenance-time (M_{maxct}), 95th percentile, not exceeding 1.5 hours when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, printed circuit boards (PCBs) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. The equipment is considered to be in a corrective maintenance mode whenever a failure or malfunction occurs, including a malfunction that allows degraded operation.

3.2.4.2 Subassembly (module) maintainability. Repairable modules shall have a M_{ct} of 2.0 hours and a M_{maxct} of 4.0 hours (95th percentile) at the organizational, intermediate, or depot level maintenance, as appropriate. M_{ct} for module repair includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all maintenance tasks. For purposes of maintainability, a module is considered the lowest replaceable subassembly (printed circuit (PC) card) for which repair is accomplished by the replacement of piece parts.

3.2.4.3 Modular construction. Predeveloped modules from the Standard Electronic Modules (SEM) program module listing of MIL-STD-1378 may be used to conform to the requirements of this specification. Procuring activity approval shall be obtained before designing a SEM module in accordance with MIL-STD-1378 for use in the equipment.

3.2.5 Environmental conditions. The equipment shall be designed to conform to the environmental requirements of 3.2.5.1 and 3.2.5.2.

3.2.5.1 Temperature. Unit 1 and Unit 2 shall operate under the temperature ranges specified for Range 4 equipment in MIL-E-16400.

3.2.5.2 Noise. The equipment shall conform to the airborne and structureborne noise acceptance limits specified in MIL-STD-740.

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3.3 Design and construction.

3.3.1 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.3.1.1 Connectors. All RF connectors utilized on the switching matrix shall be functionally equivalent to Type N as specified in MIL-C-39012 but having a 1.97 cm (0.75 in) square mounting flange, such as Amphenol connector, Part No. 9575. Connectors used for the interconnecting control and interconnectors shall be furnished. The multipin connectors used to interface the switch matrix harness with the actuator assemblies shall be a 19 ITT Cannon double density connector, derived from the MIL-C-24308 series of miniature rack and panel connectors in accordance with the Appendix to this specification. Special tools (see 6.4) required to assemble or disassemble the double density connector, such as crimping tool and locator, insertion and extraction tool, shall be supplied as part of each equipment.

3.3.1.2 Exterior finish. The exterior finish of the equipment shall conform to the requirements of MIL-E-15090.

3.3.1.3 Enclosure. The equipment shall conform to the dripproof requirements (0 to 45 degrees) specified in MIL-STD-108.

3.3.1.3.1 RF switch matrix. The switch matrix shall be capable of being mounted in a standard 48.26 cm (19 in) rack opening, a maximum internal rack width of 43.75 cm (17.50 in) (including rack mounts). The switch matrix shall be designed for ease of service and maintenance as defined in MIL-STD-189. Cabinet slides shall be provided and shall be secured to vertical rails in the rack using either contractor supplied right angle brackets or the mounting holes in the cabinet section of the slide.

3.3.1.3.2 Remote control unit. The remote control unit shall be capable of being rack-mounted in a standard 48.26 cm (19 in) rack, a maximum internal rack width of 43.75 cm (17.50 in) (including rack mounts). The 48.26 cm (19-in) mounting shall be in accordance with MIL-STD-189. The remote control unit shall be designed for ease of service and maintenance. A 62.5 cm (25 ft) cable shall be supplied as part of the remote control unit for use in connecting this unit to the RF switch matrix.

3.3.1.4 PCBs, connectors, and integrated circuits. PCBs shall be in accordance with MIL-STD-275. Printed circuit connectors shall be in accordance with MIL-C-55302. Integrated circuits shall be in accordance with MIL-E-16400 and shall be soldered to the PCB.

3.3.1.5 Material flammability. Material used as part of the transmitter antenna switching matrix group shall conform to Requirement 3 of MIL-STD-454.

3.3.2 Electromagnetic interference (EMI) characteristics. The switch matrix and remote control unit shall be self-compatible and shall conform to the requirements of MIL-STD-461 for Class IC equipments, with the following exceptions:

a. CE01 broadband is not required
b. CE02 and CE04 are not required
c. CS01: See FIGURE 3 for limits
d. CS06 shall be performed at 400 volts peak, 5 microseconds(μs) wide
e. RE01 and RS01: See FIGURE 4 for limits
f. RS03 shall be performed at 10 volts per meter

If line-to-ground powerline filters are used to conform to the MIL-STD-461 requirements, the total capacitance shall not exceed 0.1 microfarad (μF) for 60 Hz equipments, or 0.01 μF for 400 Hz equipments.

3.3.2.1 Powerline filter capacitors. If line-to-ground powerline filters are used, the total capacitance shall not exceed 0.01 μF.

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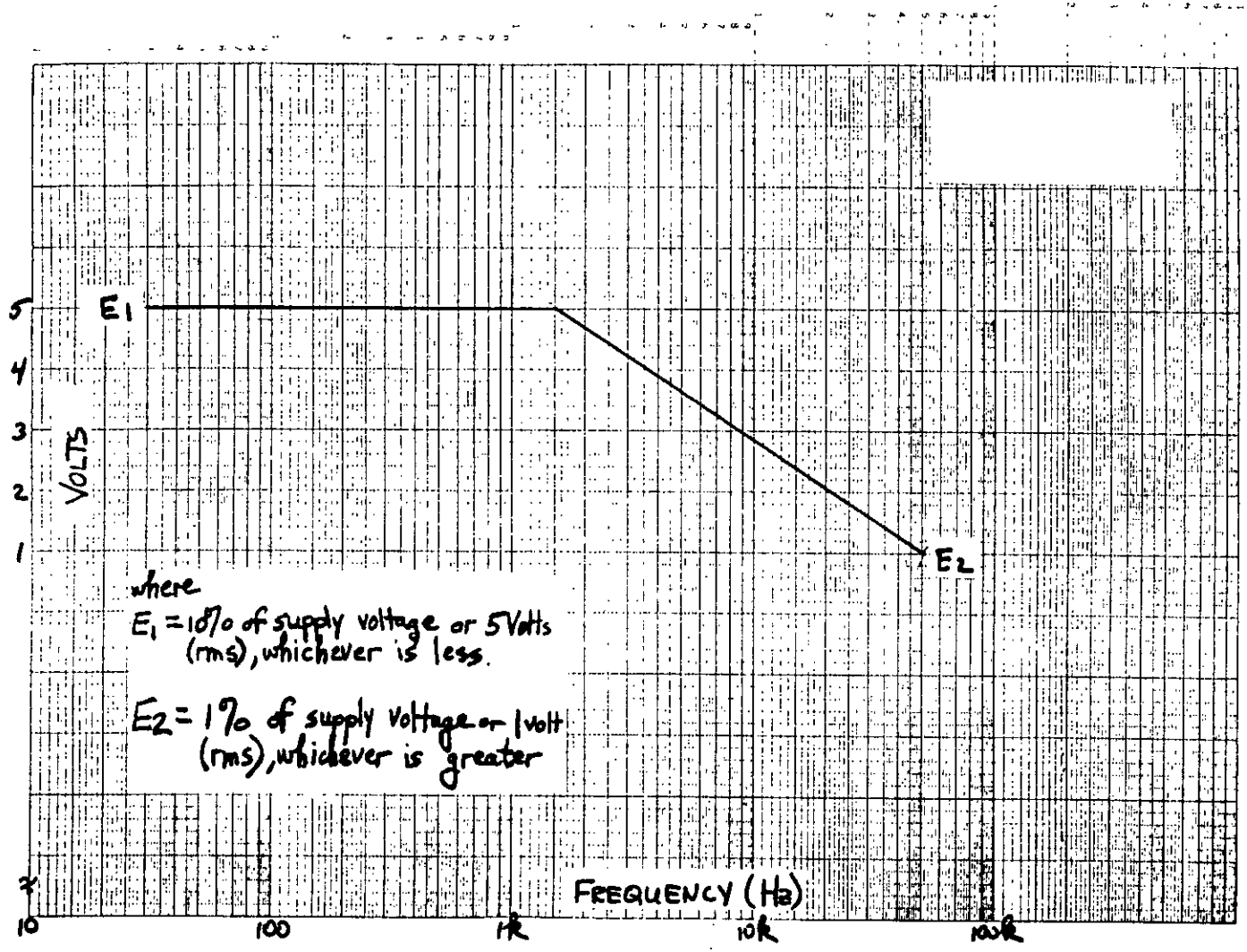


FIGURE 3. Limits for CS01.

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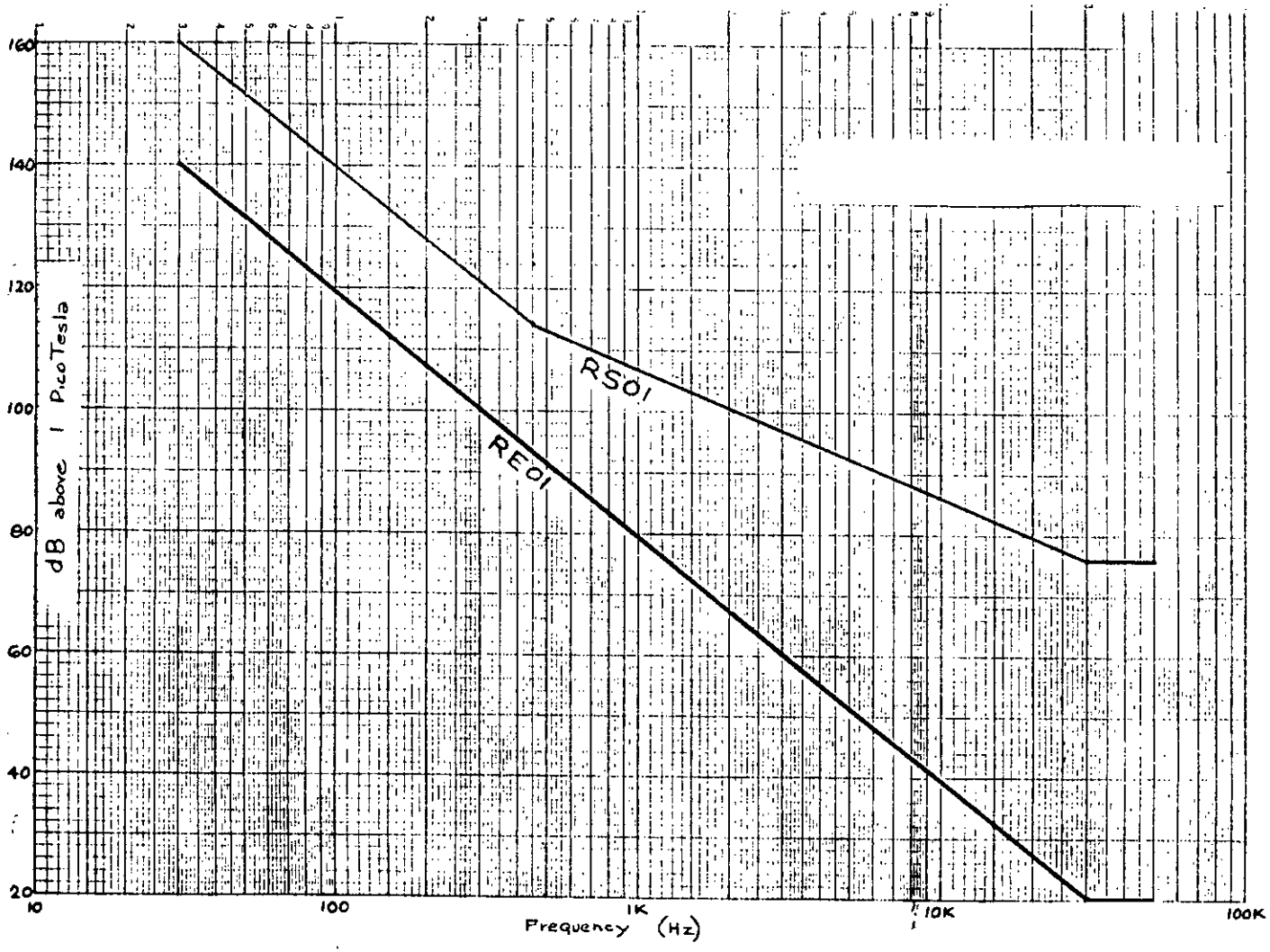


FIGURE 4. Limits for RE01 and RS01.

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3.3.2.2 Switching transients and induced noise. Switching transients from one circuit shall not exceed 7 decibels refered to 1 watt (dBW) peak induced into any other circuit, and power supply related noise induced into any circuit shall not exceed 7 decibels referred to 1 milliwatt (dBm) peak.

3.3.3 Workmanship.

3.3.3.1 Workmanship screen. All equipment shall withstand a defect detection vibration screen of random type vibration at $0.04g^2/Hz \pm 3dB$ from 80 to 350 Hz.

3.3.3.2 Temperature cycling. All equipment shall withstand a temperature cycling screen to ensure elimination of initial failures due to random defective components and marginal operation due to component lot variations or poor assembly processes and procedures.

3.3.4 Interchangeability. Unit switches of the RF switch matrix shall be replaceable and interchangeable.

3.3.5 Leakage current. The leakage current of the equipment shall be less than 5 milliamperes (mA).

3.3.6 Human engineering. The design of the equipment shall be in accordance with the requirements of MIL-STD-1472.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
    1. Production inspection (see 4.4.1)
    2. Production control inspection (see 4.4.2)
    3. Environmental inspection (see 4.4.3)
c. Reliability testing (see 4.6)
d. Maintainability demonstration (see 4.7)
e. Inspection of preparation for delivery (see 4.8)

4.3 First article inspection. Unless otherwise specified (see 6.2.1), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE II.

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TABLE II. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination						
Weight	3.2.2.1	4.5	X	X		
Size	3.2.2.1	4.5	X	X		
Parts control	3.3.1	4.5	X	X		
	3.3.1.4					
Finish	3.3.1.2	4.5	X	X		
Marking	3.1	4.5	X	X		
Safety	3.1	4.5	X	X		
Workmanship (general)	3.3.3	4.5.2	X	X		
Performance						
Split matrix design	3.2.1.11	4.5.1	X	X		
Characteristic impedance	3.2.1.5	4.5.1	X			
VSWR	3.2.1.9	4.5.1	X			
Primary power source	3.2.1.16	4.5.1	X	X		
Average power rating	3.2.1.?	4.5.1.1	X			
Peak power rating	3.2.1.3	4.5.1.2	X			
Insertion loss	3.2.1.4	4.5.1	X			
Intermodulation	3.2.1.8	4.5.1	X			
Crosstalk	3.2.1.6	4.5.1	X	X		
Isolation	3.2.1.7	4.5.1	X	X		
VSWR	3.2.1.9	4.5.1	X	X		
Interlocks	3.2.1.13	4.5.1	X	X		
Temperature						
Low temperature	3.2.5.1	4.5	X			X
High temperature	3.2.5.1	4.5	X			X
Humidity	3.1	4.5	X			X
Workmanship screen	3.3.3.1	4.5.2.1	X	X		
Interchangeability	3.3.4	4.5.3	X			X
Thermal survey	3.1	4.5	X			X
Shock	3.1	4.5	X			X
Vibration	3.1	4.5	X			X
Inclination	3.1	4.5	X			X
Airborne and structureborne noise	3.2.5.2	4.5	X			X
Enclosure	3.3.1.3	4.5				
Steady-state voltage and frequency	3.2.1.16	4.5	X			
Transient voltage	3.2.1.16	4.5	X			
Transient frequency	3.2.1.16	4.5	X			
Spike voltage	3.2.1.16	4.5	X			
Power interruption	3.2.1.16	4.5	X			
Power and power factor	3.2.1.16	4.5	X			
Leakage current	3.3.5	4.5.4	X		X	
Electromagnetic interference	3.3.2	4.5.5	X			X
Temperature cycling	3.3.3.2	4.5.2.2	X	X		
Reliability test and demonstration	3.2.3	4.6	X			
Maintainability demonstration	3.2.4	4.7	X			

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4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4 and as shown in TABLE II.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on each equipment offered for delivery. This inspection shall comprise such examination and tests as will prove the workmanship and reveal omissions and errors of the production process, such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of control adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests specified in TABLE II (Group A).

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified herein. This inspection shall consist of the examinations and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of the performance with other equipment in a system. These tests shall be performed on the complete equipment as offered for delivery. Production control inspection shall include the examinations and tests specified in TABLE II (Group B).

4.4.2.1 Sampling plan for Group B inspection. Equipments submitted for Group B sampling tests shall be selected by the Government Quality Assurance Representative and shall be representative of current production. Group B inspection, including sampling, shall conform to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall be performed on inspection lots that have passed Group A inspection. The acceptable quality level (AQL) shall be 6.5 percent defective and the inspection level shall be S-3 for normal, tightened, and reduced inspection.

4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot, using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis as specified in 4.4.3.1. This inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole; life tests, simulated service tests; test of the effects of changes of environment (such as extremes of temperature and humidity, effect of salt air); and tests of the effects of shock, vibration, and inclination. Environmental inspection shall include the examinations and tests specified in TABLE II (Group C).

4.4.3.1 Sampling for environmental inspection. One equipment shall be selected from each successive 50 equipments, or fraction thereof, produced. The first sample equipment shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Quality Assurance Representative the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. The equipment shall then be retested.

4.4.4 Reinspection of conforming production control and environmental test units. Unless otherwise specified in the contract (see 6.2.1), sample units which have been subjected to and have passed production control or environmental inspection may be accepted on the contract, provided they are resubjected to and pass the production inspection specified in 4.4.1 after repair of all visible damage.

4.5 Test methods. Unless otherwise specified herein, the examinations and tests listed in TABLE II shall be performed in accordance with the test methods specified in MIL-E-16400.

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4.5.1 Performance test. The equipment shall be subjected to a performance test including all operating controls, to determine compliance with the performance parameters and safety requirements specified herein. The performance tests shall include, as a minimum, the performance tests specified in TABLE II.

4.5.1.1 Average power rating. Each RF switch crosspoint shall be subjected to average (heating) power of 2.0 kW at any frequency from 2.0 to 30.0 MHz and 400 watts at any frequency from 225 to 400 MHz for 1 hour, continuously. During the test the switch crosspoint shall be terminated in a 50-ohm load not to exceed a 1.2:1 VSWR. After the test, the insertion loss, crosstalk, and isolation shall be measured.

4.5.1.2 Peak power rating. Each RF switch crosspoint shall be subjected to peak powers of 5.0 kW at any frequency from 2.0 to 30 MHz and 1000 watts at any frequency from 225 to 400 MHz for 1 hour, continuously. During the test the switch crosspoints shall be terminated in a load VSWR of 4:1 such that the peak voltages are induced at the switch crosspoint. After the test, the insertion loss, crosstalk, and isolation shall be measured.

4.5.2 Workmanship (general). The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process for conformance to the requirements of 3.3.3.

4.5.2.1 Workmanship screen. Prior to conducting temperature cycling, vibration shall be performed on each equipment. The vibration shall be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration.

4.5.2.1.1 The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of ten minutes on the axis deemed most susceptible to vibration excitation. All test items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 5. The control accelerometer shall be located next to one of the mounting points of the item under test. Equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density.

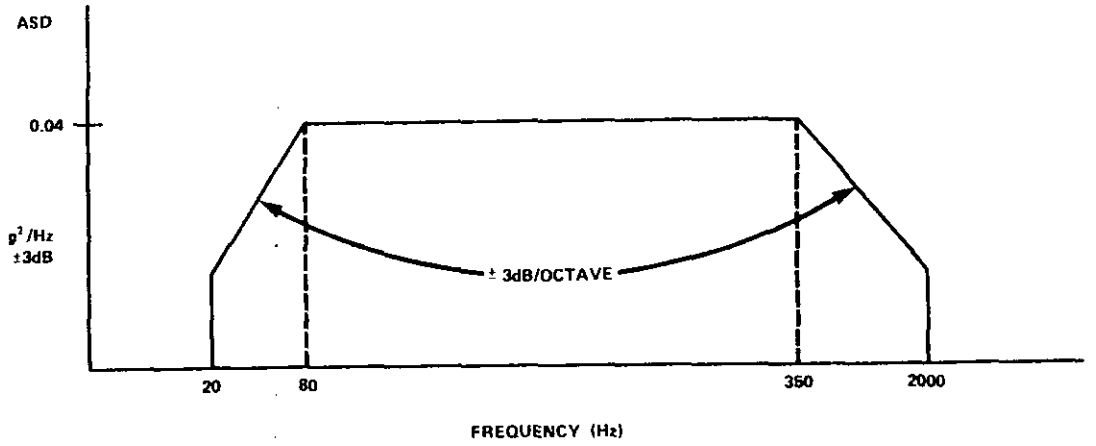


FIGURE 5. Random vibration curve.

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4.5.2.1.2 The test item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during test shall be corrected and the test repeated.

4.5.2.2 Temperature cycling. Each equipment shall be subjected to the temperature curve shown in FIGURE 6. The number of cycles required shall be determined by the electrical and electronic parts count and the parts count chart shown in TABLE III. The temperature rate of change shall not be less than 5° Celsius (C) per minute. Equipment power shall be turned on and off at the indicated times. When practical, the equipment drawers, panels, and enclosures shall be opened or removed for maximum exposure to the changing temperature. The last temperature cycle shall be failure-free. The dwell time shall be 80 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the nonoperating condition.

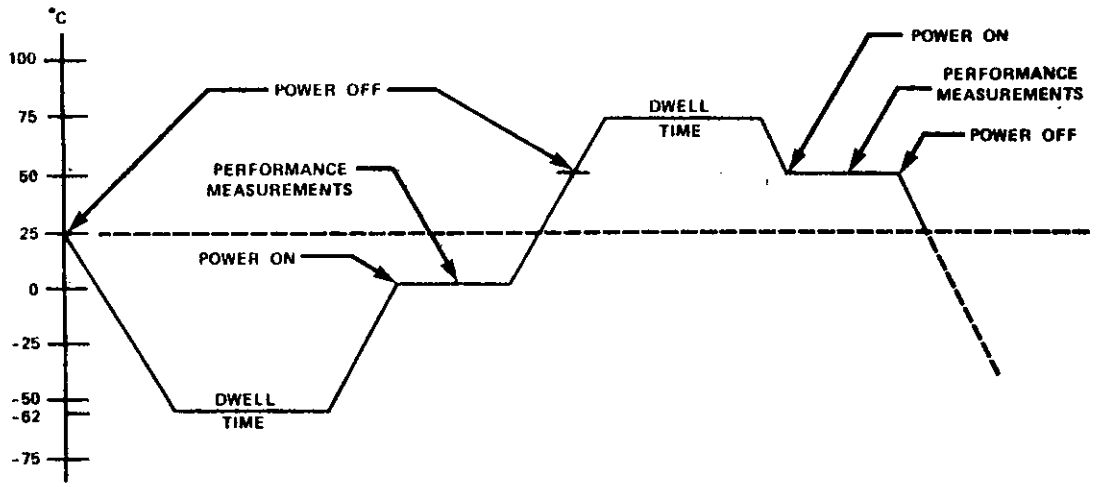


FIGURE 6. Temperature curve (one cycle).

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TABLE III. Parts count chart.

Electrical and electronic parts count	Number of temperature cycles
500 or less	3
501 to 1000	4
1001 to 2000	6
2001 to 3000	8
3001 and up	10

4.5.3 Interchangeability. The requirements for interchangeability specified in 3.3.4 shall be demonstrated on two units chosen at random from each lot of 50 or less.

4.5.4 Leakage current test.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.4.1 Equipment connections. The equipment, consisting of the RF switch matrix and remote control unit, shall be placed on an insulated surface. The safety ground conductor between the equipment and the source power shall be opened during the test. The equipment shall be connected as shown in FIGURE 7.

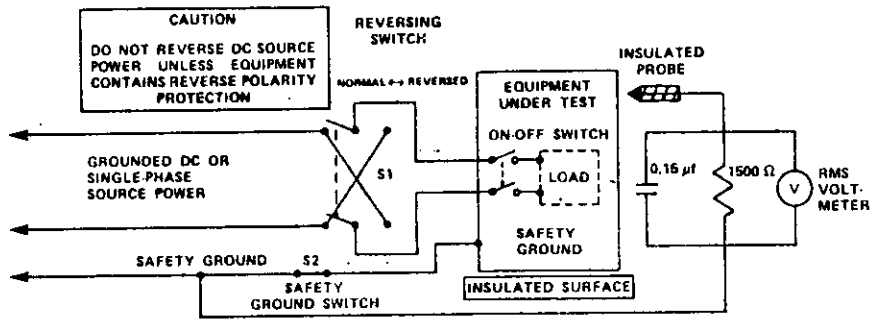
4.5.4.2 Measurement. Leakage current shall be measured on the equipment in its normal operating configuration and under maximum RF loading conditions. Equipment controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 volts at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the equipment is designed. The overall measurement error shall not exceed five percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in the test diagram. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.5 EMI test. The requirements of 3.3.2 shall be verified by tests performed in accordance with MIL-STD-462. For Group C, only tests CE03 and RS03 shall be performed.

4.6 Reliability testing. Reliability testing shall be performed in accordance with MIL-STD-781 Test Plan IVC. Test temperature shall be maintained at 50° C. Vibration stress (continuous) shall be as specified in the combined environments for shipboard equipment paragraph of MIL-STD-781. Test conditions and parameters to be measured during test shall be as follows:

a. The electrical switching load shall consist of the following:
   1. 50-ohm input over a frequency range of 2 MHz to 400 MHz
   2. Average power 2 kW over the HF band and 400 watts over the UHF band
   3. Peak power of 5 kW over the HF and 1 kw over the UHF band
b. The frequency of switching shall be 50,000 cycles with 30 seconds per position
c. One cycle shall be designated as the sequence of stepping through all positions of the switch

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GENERAL ORDER OF TEST:

1. SOURCE POWER OFF. CONNECT EQUIPMENT PER DIAGRAM.
2. ON OFF SW OFF. S1 SW NORMAL. S2 SW CLOSED CONNECT SOURCE POWER.
3. OBSERVE WARNING S2 SW OPEN. ON-OFF SW ON.
4. FOR EACH PROBE POINT, RECORD VOLTMETER READING (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. REPEAT STEP 4.
6. OBSERVE CAUTION S1 SW REVERSED. ON OFF SW ON. REPEAT STEP 4.
7. ON OFF SW OFF. REPEAT STEP 4.
8. S2 SW CLOSED S1 SW NORMAL.
9. REPEAT STEP 3 THRU 8 FOR EACH MODE OF OPERATION.
10. REMOVE SOURCE POWER. DISCONNECT EQUIPMENT. SIGN RECORD SHEET.

WARNING

DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

FIGURE 7. Single-phase or DC test diagram for leakage current measurement.

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4.7 Maintainability demonstration.

4.7.1 Maintainability equipment demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be performed by qualified technicians and shall be used to verify that the equipment M_{ct} requirements are met.

4.7.2 Demonstrated conditions. The equipment maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual PCBs) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. Sixty candidate corrective maintenance tasks shall be identified (selected in accordance with Appendix A of MIL-STD-471). The following information shall be made available for each candidate task:

a. Designation of specific faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part or simulation thereof)

The procuring activity or its authorized representative shall use the candidate tasks as a guide to select a sample of 20 tasks for the demonstration.

4.7.3 Subassembly (module) maintainability. The module maintainability requirements shall be demonstrated by replacement of piece parts at the organizational, intermediate, or depot-level as appropriate. A minimum of five faults per repairable module, but no less than 60 faults, shall be selected. Fault selection and simulation for each repairable module shall be in accordance with the procedure for the equipment maintainability demonstration. A total of 20 faults shall be selected by the procuring activity for the module maintainability demonstration.

4.7.4 Accept or reject criteria.

4.7.4.1 Corrective maintenance. The accept or reject criteria for the demonstration of the corrective maintenance times for the demonstration shall be as shown in TABLE IV.

TABLE IV. Acceptance criteria.

	M_{ct}	M_{maxct}
Acceptance level	8	0
Sample size	20	20

Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value of each specified index (M_{ct}, M_{maxct}) is less than or equal to that shown in TABLE IV corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s) and recorded as greater than or lesser than each index. The actual value shall also be recorded. An accept decision can only be made when an accept decision is made for both M_{ct} and M_{maxct}. Accept or reject criteria for preventive maintenance shall be the capability of performing each preventive maintenance task with no degration of system performance. If degradation of system operations exists, the preventive maintenance test will have resulted in a reject decision.

4.7.5 Preventive maintenance. The demonstration shall qualitatively demonstrate that on-line preventive maintenance for the equipment shall not interfere with on-line operation.

4.7.6 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the equipment will be installed.

4.7.7 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the maintainability demonstration.

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4.7.8 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation delivered with the equipment.

4.7.9 Rejection. Failure to conform to any of the requirements specified herein shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activit shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies
b. Upon approval of a documented approach, the contractor shall repeat the demonstration until an accept decision is reached.

4.7.10 Acceptance. The maintainability demonstration shall exhibit that the equipment conforms to the accept criteria prior to any delivery.

4.8 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance to the requirements of Section 5.

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for use in submarines.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. When reinspected conforming production control and environmental test units may not be accepted (see 4.4.4)
d. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified herein will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified herein will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.4	Procedures, Test	UDI-T-22710
4.4	Test Report	DI-T-5247
4.5	Report, Thermal Survey	DI-R-7036
4.5.2.2	Report, Burn-In Test	DI-R-7040
4.6	Reports, Reliability Test And Demonstration	DI-R-7034
4.6	Procedures, Reliability Tests	DI-R-7035
4.7	Plan, Maintainability Demonstration	DI-R-2129
4.9	Report, Test, Electromagnetic Interference/Electromagnetic Compatibility (EMI/EMC)	DI-R-2066/R021

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the DAR. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, tests, and approval of the first article.

6.4 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catolog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

6.5 Failure. Failure is defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements specified herein.

Preparing activity:
NAVY-EC
(Project 5820-N734)

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APPENDIX

DOUBLE DENSITY CONNECTOR

10. SCOPE

10.1 Scope. This appendix covers the requirements for the double density connector.

20. APPLICABLE DOCUMENTS

20.1 Government documents. The following documents, of the issue listed in the Department of Defense Index of Specifications and Standards (DoDISS) and its supplements, form a part of this document to the extent specified herein. The date of the applicable DoDISS and supplements thereto shall be as specified in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-395	Plastic Molding (And Extrusion) Material, Nylon, Glass Fiber Reinforced

MILITARY

MIL-M-14	Molding Plastics And Molded Plastic Parts, Thermosetting

(Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 The double density connector shall be in accordance with MIL-C-24308, except as follows:

a. Insert paragraph of MIL-C-24308: Insert material shall be nylon in accordance with L-P-395 in lieu of MIL-M-14, black color
b. Solder contacts paragraph of MIL-C-24308: Standard shall be crimp pot but may have crimped-in or integral solder pots
c. Insert design and construction paragraph of MIL-C-24308: Contact style shall be opposite to MIL-C-24308, socket contact protrude rather than the pin. Also, no metal clips, integral retention collar molded in insulator.
d. Insert arrangement paragraph of MIL-C-24308: Insert arrangements for 2DD shall be 19-, 31-, 52-, 79-, and 100-contact positions.
e. Interchangeability paragraph of MIL-C-24308: Not interchangeable.
f. TABLE I: Shall be 6.75 kg (15 lb) maximum insertion and 9.0 kg (20 lb) maximum removal, in lieu of 1.4 kg (3 lb) and 1.8 kg (4 lb), respectively
g. TABLE II: Mating and unmating force: Size 22 (5 ampere) contact only. The shell sizes shall be 19, 31, 52, 79, and 100 in lieu of 1, 2, 3, 4, 5, and 6

	Unmating				Mating	
Shell size	Minimum		Maximum		Maximum	
	(kg)	(lb)	(kg)	(lb)	(kg)	(lb)
19	0.34	0.75	2.7	6.0	4.5	10.0
31	0.45	1.00	4.5	10.0	9.0	20.0
52	0.79	1.75	7.7	17.0	13.5	30.00
79	1.12	2.50	10.8	24.00	20.3	45.00
100	1.5	3.25	13.5	30.00	24.8	55.0

h. Contact resistance paragraph of MIL-C-24308: Only contact size 22 applies
i. Contact engagement and separation forces paragraph of MIL-C-24308: Only contact size 22 applies
j. Fluid immersion paragraph of MIL-C-24308: Will not pass fluid immersion
k. Rear periphery contacts shall be marked only with molded raised numbers.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-S-23854

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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